BindCraft-Guided *de novo* Design of Peptide Binders Targeting the Bacterial Quorum-Sensing Regulator Rgg3

James Zhu[1] · Xiangqian Shi[2] · Miles D. Woodcock-Girard[3] · Arumay Pal[2] · José A. Villegas[2] · Kevin Drew[3]

[1] Pomona College, Claremont, CA, USA
[2] Department of Pharmaceutical Sciences, Retzky College of Pharmacy, University of Illinois Chicago, Chicago, IL, USA
[3] Department of Biological Sciences, University of Illinois Chicago, Chicago, IL, USA

Rosetta Commons

UIC UNIVERSITY OF ILLINOIS CHICAGO

01 ABSTRACT

THE STORY IN 10 SECONDS

CORE PROBLEM
Rising antibiotic resistance motivates anti-virulence strategies that suppress pathogenicity without inhibiting bacterial growth.

BIOLOGICAL TARGET
Rgg3 regulate *Streptococcus* virulence through short hydrophobic peptide (SHP) quorum-sensing signals.

PRIOR FOUNDATION
Earlier short helical peptide binders inhibited Rgg3 signaling, and a peptide–Rgg3 crystal structure revealed the binding interface.

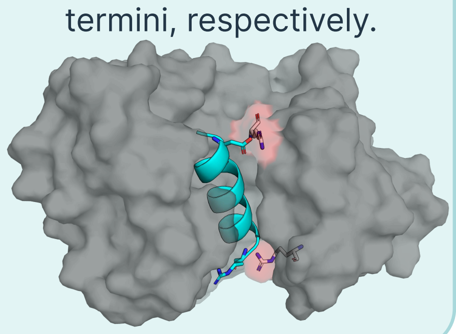
SECOND-ROUND DESIGN
The experimental Rgg3 structure guided BindCraft *de novo* design. Thousands of trajectories were ranked by structural confidence, interface quality, and sequence properties.

CANDIDATE SELECTION
The top six peptide designs were selected for functional assays of Rgg3-mediated quorum-sensing inhibition.

WHY IT MATTERS
This iterative AI–experimental workflow offers a scalable route to specific anti-virulence therapeutics against *Streptococcus* infections.

02 METHOD

Arg159 and Arg84 form salt-bridge hotspots at the native SHP N- and C-termini, respectively.



- Rgg3 (PDB 10LB)
- Original SHP (10LB)
- **Key:** Arg84 + Arg159

1. **TARGET SELECTION**
   PDB 10LB
   CHAIN A
   Residues 74–end

   HOTSPOT for 4 BATCHES:
   - 84, 152, 159
   - 84, 156, 272, 279
   - None (batches 3–4)
2. **HALLUCINATION**
   Generate candidate binder backbones and sequences around the selected Rgg3 interface.
3. **SolMPNN REDESIGN**
   Redesign non-surface residues. One batch used predict_initial_guess = true and num_seqs = 30.
4. **ALPHAFOLD2 VALIDATION + FILTERING**
   Validate candidates, retain designs passing all filters, and advance the top 6 to functional assays.

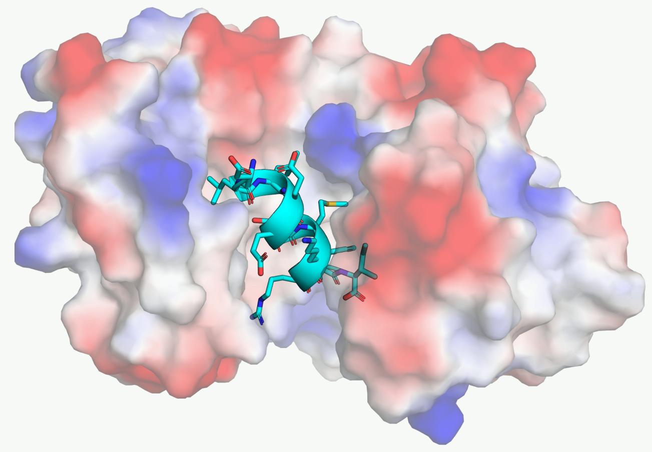
03 DESIGNS / RESULTS

Electrostatic surface · Designed peptide
Rgg3 RMSD (Å): 6W1F inactive / 7JI0 active · Rgg3 Arg84

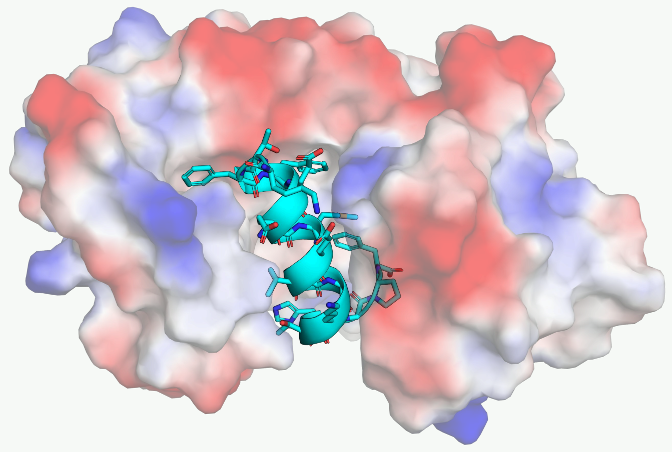
COMPUTATIONAL SELECTION FUNNEL

2,500+ TRAJECTORIES → **565** PASSED HALLUCINATION FILTERS → **93** PASSED MPNN FILTERS → **6** PASSED CUSTOMIZED FILTERS

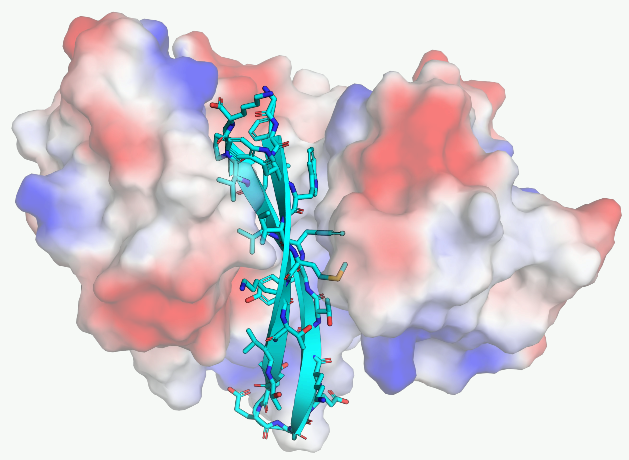
10LG_I11_s874366_mpnn1
RMSD: 0.651 / 2.078 Å



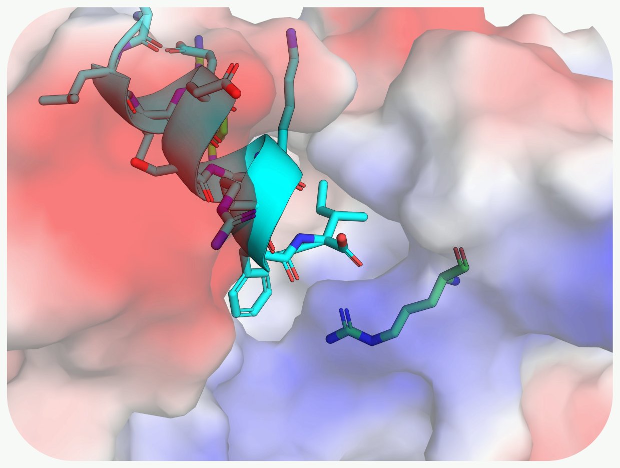
10LG_I18_s756121_mpnn1
RMSD: 0.741 / 2.359 Å



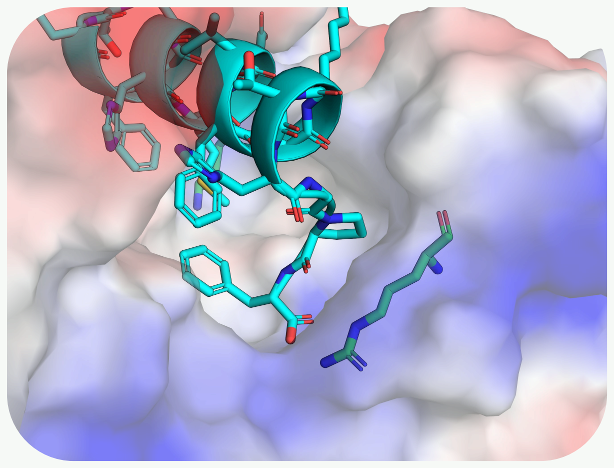
10LG_I25_s62213
RMSD: 0.861 / 2.456 Å



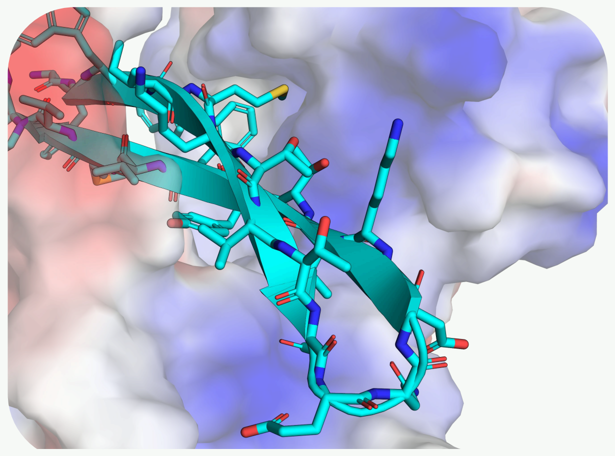
Zoom-in: 10LG_I11_s874366_mpnn1



Zoom-in: 10LG_I18_s756121_mpnn1



Zoom-in: 10LG_I25_s62213



RGG3 DESIGNS SELECTED FOR EXPERIMENTAL EVALUATION

design_name	sequence	i_pTM	surface_hydrophobicity	shape_complementary	GRAVY*
10LG_I22_s178631_mpnn2	DITKEQWWKYGPMGAWWKVIGH	0.92	0.64	0.68	-0.768
10LG_I17_s37923	KIKTPFEVAQEHFLTKT	0.92	0.44	0.67	-0.541
10LG_I11_s874366_mpnn1	DLLEESMKRFI	0.91	0.45	0.64	-0.264
10LG_I18_s756121_mpnn1	KTFFEEHMEVFGKTHGPF	0.90	0.44	0.66	-0.622
10LG_I14_s250321	NPLDITMKRWHTEV	0.90	0.43	0.66	-0.829
10LG_I25_s62213	GFWWIMKTVTDENGEKVTYFVMVPK	0.90	0.60	0.66	-0.104

*(Grand Average of Hydropathy)

04 CONCLUSION

1. Rgg3's buried hydrophobic pocket caused filter failures, although similar short lab peptides remained soluble.
2. The tightly packed binding site limited mutation, so SolMPNN produced minimal redesign.
3. Design trajectories showed a strong preference for methionine, causing the designs fail to pass the final filters.

FUTURE DIRECTIONS
- Test the six candidates
- Restrict methionine during hallucination
- Penalize interfaces with >3 Lys + Met

05 ACKNOWLEDGEMENTS

Thanks to all the members of the Drew Lab and Villegas Lab. This work was supported by the National Science Foundation under Award No. 2548392 through the Rosetta Commons REU Program.

Trajectory Data

MEET JAMES

I'm James Zhu, a Math and Physics student at Pomona College applying machine learning, molecular simulation, and high-performance computing to protein design and peptide biophysics.

CONNECT ON HIHELLO & LinkedIn
jzii2024@mymail.pomona.edu